Learning-Based Detection of Smoke-Degraded Regions in 3D LiDAR Scans

Thesis Proposal – Data Science MSc., TU Wien

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1 Introduction

Most autonomous robots in safety-critical indoor environments (e.g., industrial facilities, burning buildings) rely on 3D LiDAR for localization and mapping while operating in smoke, dust, and other visual degradations [2]. In smoke-filled environments, LiDAR returns can become sparse or distorted due to scattering and attenuation, which leads to degenerate scan geometry and failures in downstream state estimation [7].

Existing sensor fault detection in robotics is mostly model-based (e.g., multiple-model Kalman filtering for internal sensor faults) and assumes known failure modes, so it does not address environment-induced effects like smoke [10]. Recent robust LiDAR odometry and fusion methods detect degeneracy using geometric or overlap criteria inside specific SLAM/odometry pipelines, such as adaptive LiDAR-radar fusion in dense smoke or degradation-aware LiDAR-radar-inertial odometry, but they do not expose a general, reusable reliability module [8]. Atmospheric LiDAR work shows that smoke can be detected from LiDAR signals using simple statistical features and ML on vertical backscatter profiles, [9] and LiDAR anomaly-detection methods for autonomous driving use feature-based detectors or variational autoencoder (VAE) models for generic anomalies rather than calibrated, per-region reliability estimates in indoor smoke scenarios [11, 13].

Goal. Design, train, and evaluate neural networks that predict LiDAR reliability degradation due to smoke. The models will be trained on data from an indoor environment scanned first without smoke and then with smoke, and deployed as a standalone ROS 2 node that subscribes to `PointCloud2` messages and publishes reliability maps.

2 Research Questions

This thesis investigates whether neural networks can provide reliable, smoke-aware LiDAR perception for indoor robotics, and how different modeling choices affect their performance. The work is guided by the following questions:

- To what extent can neural models detect smoke-degraded regions in 3D LiDAR scans?
- How can such models be integrated to provide region-specific reliability estimates that are usable within LiDAR-based odometry pipelines?
- How does local processing (per-voxel features) compare to global processing (whole-scan)?
- How do temporal models (multi-scan sequences) compare to single-scan models?

3 Proposed Approach

3.1 Data Collection and Labeling

- Map an indoor environment without smoke using a 3D LiDAR to build a high-quality clean reference map.
- Introduce smoke into parts of the same environment using a smoke machine and record 3D LiDAR scans covering both clean and smoke-filled regions.
- Align each smoky LiDAR scan to the clean reference map using standard LiDAR scan-to-map registration, for example a coarse feature-based alignment followed by ICP refinement [1].
- For every point in each smoky scan, compare its position and intensity against the expected structure in the clean map along the corresponding ray (e.g., distance to the nearest mapped surface, missing expected returns, strong attenuation relative to the clean map).
- Label points with a large mismatch with respect to the clean map as *degraded*, and points that match the clean map well as *reliable*, using thresholds that are tuned on small manually inspected subsets; the bulk of label generation remains automatic.
- Optionally apply simple spatial or temporal smoothing of labels to avoid isolated mislabeled points, while keeping the labeling process largely automatic.

This per-point labeling strategy is analogous in spirit to the quasi-unsupervised smoke plume detection of Rossi et al., where smoke is identified as deviations from a learned "safe" lidar backscatter profile, [9] but here it is adapted to 3D indoor scans and map-based comparison rather than vertical profiles.

3.2 Model Design

The core of the thesis is supervised neural models that classify individual LiDAR points as reliable vs. degraded and output a reliability probability. Two complementary design dimensions will be explored: (1) local vs global processing of individual LiDAR scans, and (2) single-scan vs temporal modeling across multiple scans.

3.2.1 Single-Scan Model

Local processing (per-voxel):

- Convert `PointCloud2` to voxels (e.g., 10cm cubes) and compute per-voxel statistics (such as point density, mean intensity, local geometry).
- Apply a MLP/CNN on per-voxel features to calculate per-voxel reliability.

Global processing (whole scan):

- Feed the full `PointCloud2` scan into a 3D CNN or transformer.
- Predict per-point reliability using global scan context.

Vertical LiDAR smoke-detection work already demonstrates that simple statistics over range–time backscatter, combined with ML classifiers, can robustly identify smoke. [5, 9] The single-scan model transfers this idea to 3D indoor point clouds.

3.2.2 Temporal Model

Smoke-induced degradation often exhibits temporal structure (growth, movement, clearing). To exploit this both local and global processing will be extended to sequences of T scans:

- **Local temporal**: track voxels across scans and apply a lightweight recurrent neural network or temporal convolutional network on voxel feature sequences.
- **Global temporal**: feed a sequence of full scans into a video transformer or temporal convolutional network.

Unsupervised VAE-based anomaly detection on spatio-temporal LiDAR has been shown to benefit from capturing both spatial and temporal patterns [11]. Here, temporal structure is used in a supervised setting targeted specifically at smoke-induced degradation.

3.3 ROS 2 Node

The node subscribes to LiDAR topics (e.g., `/points` carrying `PointCloud2` messages) and, if needed, to pose or time-synchronization topics. For each incoming scan, it runs the model to obtain reliability probabilities and publishes the per-point/per-voxel reliability.

The node encapsulates the full pipeline from preprocessing and feature extraction to model inference, aggregation, and message formatting. It is intended to be agnostic to any specific SLAM or control stack, so it can be reused in any ROS-based system where smoke-induced LiDAR degradation is relevant.

4 Evaluation Plan

4.1 Point- and Voxel-Level Detection Metrics

- Train / validation / test split on the indoor smoke dataset.
- Metrics at the point or voxel level:
  - ROC and PR curves, AUC, F1-score.
  - Calibration plots (e.g., reliability diagrams, Brier score).
- Compare local vs. global processing, as well as single-scan vs. temporal models.

4.2 Qualitative Analysis

- Visualize reliability maps overlaid on 3D point clouds:
  - Check that areas heavily affected by smoke (visually) are assigned low reliability.
  - Verify that stable, unobscured structures are treated as reliable.
- Compare reliability maps across time for sequences where smoke moves in and out of the field of view.

5 Literature Overview

This section lists representative works and how they inform specific design choices.

5.1 Robust LiDAR Odometry and Fusion under Degradation

- **Noh & Kim (2024).** [8] Adaptive LiDAR–radar fusion across dense smoke uses overlap-based degeneracy detection to reweigh LiDAR. This motivates the need for a reliability signal but remains heuristic and tightly coupled to a particular fusion design.
- **Nissov et al. (2024).** [7] Degradation-resilient LiDAR–radar–inertial odometry uses degradation-aware fusion. It demonstrates the benefits of reliability-aware fusion, but the degeneracy logic is embedded in the odometry pipeline.
- **Liao et al. (2024).** [4] Real-time degeneracy sensing and compensation for LiDAR SLAM using geometric clustering (DBSCAN). This is a strong non-ML baseline for degeneracy sensing but does not provide a learned per-region probability.
- **Marmaglio et al. (2025).** [6] Degeneracy-resilient LiDAR odometry via reflectance-derived correspondences. Useful as an algorithmic baseline family that mitigates degeneracy without ML; in contrast, this thesis learns a reliability score from data.

5.2 LiDAR-Based Smoke and Aerosol Detection

- **Rossi et al. (2020).** [9] Adaptive quasi-unsupervised smoke plume detection using vertical LiDAR backscatter and statistical features with an SVM. Shows that smoke is detectable from LiDAR signals using relatively simple features and motivates semi-automatic label generation via comparison to reference profiles.
- **Lolli (2023).** [5] Survey of ML techniques for vertical LiDAR-based aerosol and cloud detection. Provides a catalog of features and ML architectures that can inspire feature representations for smoke-induced degradation in 3D scans.
- **Utkin et al. (2009).** [12] Simple neural-network classifier on 1D LiDAR range profiles for forest-fire smoke detection. This is a conceptual precursor: a neural classifier on LiDAR for smoke, but with different geometry and output (binary plume detection vs. per-point reliability).

5.3 LiDAR Anomaly and OOD Detection

- **Zhang et al. (2024).** [13] Feature-based anomaly detection in LiDAR point clouds for autonomous vehicles. Demonstrates that anomaly scores on point clouds are feasible and motivates using supervised classifiers on LiDAR features.
- **Hattori et al. (2023).** [3] Anomaly detection in LiDAR data using virtual and real observations, highlighting the use of time-series data and virtual-real pairs. Supports the idea of temporal modeling and simulation-informed training.
- **Sboui et al. (2024).** [11] CNN–BiLSTM VAE for unsupervised anomaly detection in non-image LiDAR data. Serves as a strong unsupervised baseline family and justifies comparing supervised reliability estimation against VAE-style approaches.
- **Bogdoll et al. (2022).** [2] Survey of anomaly detection in autonomous driving. Provides the broader anomaly/OOD detection context and clarifies where this thesis sits (supervised, task-specific reliability estimation on LiDAR).

6 Expected Outcomes and Scope

6.1 Expected Deliverables

- Labeled dataset of indoor 3D LiDAR scans with and without smoke, with per-point reliability labels obtained via automatic comparison to a clean map.
- A ROS 2 node that implements the reliability estimator and publishes reliability maps.
- Experimental evaluation of detection performance and ablation results (feature design, architecture, temporal vs. single-scan, voxel-based vs. global).

7 Relevance to the Data Science MSc

The thesis is closely aligned with the Machine Learning in intelligent systems aspects of the Data Science MSc:

- Supervised learning on high-dimensional sensor data, including feature design, neural architecture selection, and calibration.
- Development and comparison of several different models to find the one best suited for the given task.
- Reliability and anomaly estimation in a safety-critical context.
- Practical data engineering: real-world sensor log handling, labeling, and evaluation.
- Deployment of ML models as ROS 2 nodes, demonstrating integration of data science methods into robotics software infrastructures.

By focusing on a neural reliability estimator and a clean evaluation, and treating robotics integration as a modular interface (ROS 2 node) rather than a full SLAM/control stack, the project remains a data-science thesis with clear, well-defined engineering deliverables.

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